Smart Farm Management System

Overall Introduction

Poultry Farming is a very important sector that can provide great performance and economic benefits. However, if chicken health, feed management, egg production and environmental conditions are not properly monitored, production decreases, economic losses, and the risk of disease outbreaks increase.

This research aims to develop a trained AI model and a web-based system to provide accurate, data-driven decisions in poultry farm management. It includes 4 areas:

1. Poultry Disease Detection – A deep learning model to detect poultry diseases (Newcastle, Fowl Pox, Avian Influenza).

2. Feed Optimization & Growth Prediction – AI-based feed recommendation based on chicken growth and weight measurements.

3. Egg Production & Quality Prediction – Predicting and alerts on egg production, quality, shell strength, yolk color, etc.

4. Environment & Farm Monitoring – Temperature, humidity, ammonia levels, water quality, and ventilation environmental conditions monitoring + risk prediction + decision support.

This system can be maintained without IoT devices through manual data entry, and can later be integrated with IoT to integrate live data. Through simulation / what-if analysis, it provides the ability to predict farm events, enabling safe, profitable decisions.

Benefits:

- Improving poultry health and production
- Reducing the risk of egg and chicken growth reduction
- Minimizing economic losses
- Providing an accessible, decision-support system for micro and medium farms

This research aims to make poultry farm management more efficient, observable, and fully data-driven by integrating AI, Machine Learning, IoT simulation, and a web-based dashboard.

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1. Poultry Management and Predictive Modeling for Egg Production: Accuracy of Machine Learning Models in Predicting Chickens for Culling and Egg Grading (K.T.Samarasekara – IT22366740)

Research Question

How accurately can machine learning models predict chickens for culling and grade eggs using physical and production data?

Abstract

This research explores the application of machine learning (ML) models to improve poultry management by accurately predicting chickens for culling and grading eggs based on physical and production data. Traditional methods rely on manual inspection, which is subjective and inconsistent, leading to inefficiencies. By leveraging data such as chicken breed, age, weight, feed intake, egg size, and shell quality, ML models aim to optimize these decisions to reduce economic losses and enhance product quality. The study evaluates the predictive accuracy of various ML algorithms, demonstrating their potential to transform poultry operations into a data-driven, sustainable practice.

Introduction

The poultry industry is critical to global food security, providing essential protein through eggs and meat. Effective poultry management plays a vital role in maximizing productivity and maintaining quality standards. Manual practices in poultry farms, such as culling underperforming chickens and grading eggs based on visual criteria, are labor-intensive, error-prone, and subjective. Recent advances in artificial intelligence and machine learning present promising opportunities to automate and improve these processes. This research investigates ML-based predictive models' capacity to accurately identify chickens for culling and classify eggs, contributing to operational efficiency and economic benefits.

Research Gap

While several studies have applied machine learning techniques to poultry farming, most existing research focuses on isolated tasks, such as predicting egg production or classifying egg quality. Very few studies address both culling prediction and egg grading in an integrated system, despite their direct impact on farm profitability. Additionally, current approaches often rely on limited features or basic statistical models, ignoring the potential of multi-parameter predictive analytics and advanced ML algorithms for higher accuracy. There is also a lack of practical, real-time solutions tailored for small and medium-scale poultry farms. This gap highlights the need for a robust, data-driven system that simultaneously predicts culling decisions and grades eggs accurately, providing actionable insights for efficient poultry management

2. Poultry Disease Detection and Early Prediction (W.W.M.P.W.Y.R.W.Palamakumbura – IT22324542)

Research Question:

Can machine learning and deep learning models accurately predict the identification of healthy and sick chickens and recommend treatment for sick chickens?

In future development: Detecting diseases before they spread (at an early stage).

Multimodal Disease Detection

- Combine image data (feathers, comb, droppings) with sensor data (temperature, sound of coughing, feeding behavior).

Early-stage Disease Detection

- Train models to detect very subtle changes in chicken appearance before symptoms become severe.

Dataset Contribution

- Create a **new dataset** (e.g., high-resolution images from Sri Lankan poultry farms, including under-researched diseases).

Abstract

Poultry farming plays a vital role in global food security, yet it remains highly vulnerable to infectious diseases that can spread rapidly and cause significant economic losses. Traditional disease detection in poultry relies on manual observation of visible symptoms such as feather condition, comb color, or changes in droppings, which are often subjective and fail to identify diseases at an early stage. Recent advancements in artificial intelligence have enabled automated image-based disease detection, but most existing approaches are limited to visual symptoms and overlook subtle behavioral and physiological changes that precede visible illness.

This research proposes a **multimodal deep learning framework** for poultry disease detection, combining image data (feathers, comb, droppings) with sensor data (temperature, respiratory sounds, and feeding behavior) to achieve more accurate and robust diagnosis. In particular, the system emphasizes **early-stage disease detection** by identifying subtle anomalies before symptoms become severe, thereby enabling preventive interventions. Furthermore, the study

contributes a **novel dataset from Sri Lankan poultry farms**, consisting of high-resolution images and sensor recordings that capture under-researched diseases and local farming conditions, enhancing the generalizability of detection models.

The proposed framework will be evaluated against existing image-only approaches using metrics such as accuracy, precision, recall, and F1-score, with additional emphasis on early detection performance. Beyond disease identification, the system aims to provide actionable recommendations for farmers, such as isolating suspected birds or seeking veterinary care. By integrating multimodal data, early anomaly detection, and a new regional dataset, this research advances the state of poultry health monitoring and offers a practical solution for improving farm productivity and disease management in resource-limited settings.

Introduction

Poultry farming is one of the fastest-growing sectors in the agricultural industry, providing a vital source of protein through meat and eggs. However, poultry health management faces significant challenges due to the frequent occurrence and rapid spread of infectious diseases such as Newcastle disease, avian influenza, and coccidiosis. These diseases not only reduce productivity but also result in considerable economic losses for farmers and threaten food security. Early detection and timely intervention are critical to preventing large-scale outbreaks, reducing mortality, and ensuring sustainable poultry farming practices.

Traditional poultry disease diagnosis in farms relies on manual inspection of physical symptoms such as changes in feathers, comb, droppings, and feeding behavior. These methods are subjective, labor-intensive, and often fail to detect diseases at an early stage, when preventive measures are most effective. Moreover, resource-limited settings such as rural farms in Sri Lanka lack access to advanced veterinary services and laboratory testing facilities, further complicating timely disease detection and control.

With recent advances in artificial intelligence, particularly in machine learning and deep learning, automated disease detection systems have gained increasing attention. Existing research has primarily focused on using **image-based approaches**, where convolutional neural networks are trained to classify visible symptoms on poultry images. While these methods have shown promising results, they are limited in scope, as many diseases present subtle or non-visible symptoms during the early stages. Consequently, relying solely on image data may delay detection until the disease has progressed, by which time interventions are less effective.

To address these challenges, this research proposes a **multimodal poultry disease detection framework** that integrates image data (feathers, comb, droppings) with sensor data such as environmental temperature, feeding patterns, and respiratory sounds. By fusing visual and non-

visual information, the system aims to improve accuracy and detect subtle health anomalies before clinical symptoms become severe. Additionally, the research emphasizes **early-stage disease detection** through anomaly detection and temporal monitoring, which can provide farmers with alerts before a full outbreak occurs.

Another significant contribution of this study is the **creation of a new dataset from Sri Lankan poultry farms**, capturing both image and sensor data under local conditions. Most existing datasets are limited to specific breeds and environments outside South Asia, which restricts their generalizability. The inclusion of Sri Lankan data, including under-researched poultry diseases, addresses this gap and contributes to building more robust, context-specific models.

Ultimately, the proposed system not only seeks to classify healthy and sick chickens but also to recommend appropriate actions for farmers, such as isolation of suspected birds or seeking veterinary advice. By combining **multimodal deep learning, early-stage detection, and a novel dataset contribution**, this research aims to enhance poultry disease management, reduce losses for farmers, and contribute to food security in Sri Lanka and beyond.

Research Gaps

- **Limited Multimodal Data Integration:** While many existing studies focus on image-based disease classification using deep learning, there is a lack of comprehensive approaches that integrate multiple data modalities such as images, environmental sensors, feeding behavior, and respiratory sounds to enhance detection accuracy and early diagnosis.
- **Early-stage Disease Detection Deficiency:** Current models predominantly detect diseases based on visible symptoms at advanced stages, missing subtle early indicators. There is a clear need for models trained to identify early subtle health anomalies and pre-symptomatic changes, enabling timely intervention before large-scale outbreaks.
- **Dataset Limitations and Regional Specificity:** Most publicly available poultry disease datasets are geographically limited, lacking diversity in breed, environment, and disease types, especially in South Asia. There is a scarcity of high-resolution, context-specific datasets from Sri Lankan poultry farms that include under-researched diseases and combine sensor data with visual information.

3. Chicken Feed Optimization & Growth Prediction : (Samaraweera S.N.S – IT22115034)

- ## Introduction

Good growth of chickens and efficiency of feed consumption are the main objectives of poultry farming. It is essential for a healthy growth of a chicken flock, increasing egg production and increasing the profits of the farmer.

Generally, in most agricultural systems, the same feed plan is followed for all flocks. Therefore, the different growth patterns of each flock are not considered. The aim of this research is to design a highly efficient feed plan for each chicken flock, optimize the type and amount of feed according to the flock-specific growth pattern, and provide future growth prediction.

This research will enable farmers to provide flock-specific feed recommendations. This can provide a practical solution to maximize growth, minimize feed costs, and increase overall profit. Simply put, it is "creating a smart system that provides the best feed recommendation, growth prediction, and cost-effective feed plan for each flock."

- ## Research Gap

1: Per-flock Adaptive Feed Recommendation

- Most studies follow the same feed plan for all flocks. Therefore, flock-specific growth differences are not considered.
- **Research Gap:** Adaptive systems that consider different growth patterns of each flock and provide feed recommendations are rare.
- **Novelty:** Providing a customized feed plan by analyzing flock-specific growth patterns.

2: Time-series Growth Prediction

- Most studies use only past data points to predict growth rates. Accurate models for future growth prediction are rare.
- **Research Gap**: Models that combine chicken age + feed data + time-series analysis to predict future growth are limited.
- **Novelty:** Chicken age + feed data are combined to make future growth prediction accurate.

3: Cost-aware Feed Plan

- Most studies focus only on accuracy/weight gain metrics, with minimal feed cost/profit analysis.
- **Research Gap**: Economic optimization systems that minimize feed cost and maximize growth are rare.
- **Novelty:** Economic optimization system that minimizes feed costs and maximizes growth.

- **Research articles and resources**

1. Artificial intelligence-based metabolic energy prediction model for animal feed proportioning optimization : https://www.tandfonline.com/doi/epub/10.1080/1828051X.2023.2236132?needAccess=true
2. How to Grow Fodder Indoors for Chickens : https://www.bootstrapfarmer.com/blogs/how-to-guides/how-to-grow-fodder-for-chickens
3. 16% Protein Homemade Chicken Feed Recipe for Laying Hens {No Fish Meal!}: https://silverhomestead.com/homemade-chicken-feed-recipe-for-laying-hens/
4. Optimization Of Chicken Feed Production Using Goal Programming Method At Pt. Japfa Comfeed Indonesia Tbk Unit Makassar : https://www.researchgate.net/publication/363495227_Optimization_Of_Chicken_Feed_Production_Using_Goal_Programming_Method_At_PT_Japfa_Comfeed_Indonesia_Tbk_Unit_Makassar/fulltext/636c7b3954eb5f547cbbc8cf/Optimization-Of-Chicken-Feed-Production-Using-Goal-Programming-Method-At-PT-Japfa-Comfeed-Indonesia-Tbk-Unit-Makassar.pdf
5. Research Article Optimization of Poultry Feed Formulation Using Experimental Design Methodology : https://www.researchgate.net/profile/Marius-Somda/publication/348817739_Optimization_of_Poultry_Feed_Formulation_Using_Experimental_Design_Methodology/links/6012875392851c2d4dfc06c7/Optimization-of-Poultry-Feed-Formulation-Using-Experimental-Design-Methodology.pdf
6. Crossbreeding to improve local chicken breeds: predicting growth performance of the crosses using the Gompertz model and estimated heterosis : https://www.researchgate.net/publication/370921264_Crossbreeding_to_improve_local_chicken_breeds_Predicting_growth_performance_of_the_crosses_using_the_Gompertz_model_and_estimated_heterosis/fulltext/64a70b89c41fb852dd574220/Crossbreeding-to-improve-local-chicken-breeds-Predicting-growth-performance-of-the-crosses-using-the-Gompertz-model-and-estimated-heterosis.pdf

7. End-weight prediction in broiler growth : https://www.researchgate.net/profile/Jean-Marie-Aerts/publication/6996080_End-weight_prediction_in_broiler_growth/links/5637913608aed65d3c42c816/End-weight-prediction-in-broiler-growth.pdf

- **Datasets - Chicken Feed & Growth**

1. Inferring Broiler Chicken Weight https://www.kaggle.com/datasets/lucasheilbuthh/inferring-broiler-chicken-weight
2. Poultry Form Broiler Feeder Dataset : https://www.kaggle.com/datasets/aravindhof/broilerfeeder
3. Chicken Weight vs Chicken feed : https://www.kaggle.com/datasets/yamqwe/chicken-weight-vs-different-chicken-feeds
4. Growth analysis of chickens fed diets varying in the percentage of metabolizable energy provided by protein, fat, and carbohydrate through artificial neural network : https://www.sciencedirect.com/science/article/pii/S003257911942419X

4. Environment & Farm Monitoring (Pathirage P D V P – IT22097606)

- **Introduction**

In poultry farming, the growth, egg production and overall performance of chickens are directly affected by environmental conditions. If factors such as temperature, humidity, ammonia level, ventilation and water quality are not properly monitored, there is a risk of reduced chicken productivity, farm losses and increased economic impact.

Existing systems often use IoT sensors to provide only environmental monitoring and alerts. However, AI-based systems to predict future risks and suggest corrective actions are often lacking.

The aim of this research is to develop an AI model and a web application that can predict potential environmental risks using historical data and real-time inputs, provide actionable recommendations to farmers and provide decision support to maintain optimal farm conditions.

Research Gap

AI-based, predictive, multi-parameter decision-support system does not exist, with manual or IoT-free implementation options.

1. Current poultry farm systems mostly provide only environmental monitoring + alerts.
2. They are reactive – do not take preventive action before conditions become critical.
3. Multi-parameter analysis (temperature, humidity, ammonia, ventilation, water quality) combined risk scoring is rarely used.
4. Systems are mostly IoT dependent, not practical for small farms / resource-limited environments.
5. Economic / productivity impact (egg loss, growth reduction) is rarely linked.

Novelty

1. Predictive Analytics – AI model predicts future environmental risks before critical levels reached.
2. Decision-Support System – Suggests corrective actions based on predictions (ex: Turn on ventilation).
3. Manual & Automatic Data Entry Options – Works with manual entry now; future IoT integration possible.

4. IoT Device Simulation in Web App - Farmers can simulate IoT sensor readings (temperature, humidity, ammonia, etc.) inside the web app.This allows testing the system without physical devices by entering values manually or running a virtual sensor model.

5. IoT-Free Implementation Option – Can work without sensors, suitable for small farms.

Research articles and resources

1. Smart Chicken Farming: Monitoring System for Temperature, Ammonia Levels, Feed in chickenfarm: https://www.researchgate.net/publication/343109318_Smart_Chicken_Farming_Monitoring_System_for_Temperature_Ammonia_Levels_Feed_in_Chicken_Farms/fulltext/5f1d2447a6fdcc9626b38ab9/Smart-Chicken-Farming-Monitoring-System-for-Temperature-Ammonia-Levels-Feed-in-Chicken-Farms.pdf?origin=scientificContributions
2. A Smart Automation System for Controlling Environmental Parameters of Poultry Farms to Increase Poultry Production : https://www.researchgate.net/profile/Md-Hossain-257/publication/371915319_A_Smart_Automation_System_for_Controlling_Environmental_Parameters_of_Poultry_Farms_to_Increase_Poultry_Production/links/64ca7f4c40a524707b8de03c/A-Smart-Automation-System-for-Controlling-Environmental-Parameters-of-Poultry-Farms-to-Increase-Poultry-Production.pdf
3. Computer Vision in Chicken Monitoring System Using Machine Learning: A General Review : https://www.e3s-conferences.org/articles/e3sconf/pdf/2023/85/e3sconf_icenis2023_02014.pdf

Datasets

1. poultry conditions : https://www.kaggle.com/datasets/chelbizineb/poultry-conditions?
2. Environmental Effect on Egg Production : https://www.kaggle.com/datasets/faysal1998/environmental-effect-on-egg-production/data
3. Prediction of egg production rate in poultry: https://www.kaggle.com/datasets/deepikabidri/prediction-of-egg-production-rate-in-poultry